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IoT based smart solar energy monitoring systems

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ABSTRACT

Today our society needs more energy for day-to-day activities due to rapid globalization and industrialization. In order to minimize the stress and dependency on fossil fuel, the most sustainable way is to harness suns energy. Solar energy is characterized by low cost, environment friendly, does not require frequent maintenance and most importantly, negligible maintenance cost. However, there is a requirement of monitoring solar installation in order to maximize the output power by setting real-time angles with the suns position. This can be easily done by the adoption of IoT technologies. The Internet of Things is often utilized in the measurement of solar energy for efficiency. It's also utilized to maintain the solar plant's health. The cost of renewable energy technology is decreasing throughout the globe, which encourages the construction of large-scale solar plants. Automation of plant observation on such a large scale of preparatory deployments requires sophisticated systems reliant on Internet connections since the majority of field units are situated in isolated, remote locations and therefore are not overseen from a central office. The project is based on the use of the most up-to-date, cost-effective method for remotely monitoring a solar plant performance by the inclusion of IoT. It can assist with plant maintenance, problem diagnostics, and real-time monitoring.

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1. Introduction

Solar power facilities must be monitored for optimum electricity output. This helps to restore economic power production from power plants by replacing defective star panels, looking for contacts and alternating those issues with sludge accumulated on output reducing panels and shaking wire. However, we recommend a machine-controlled Internet of Things (IoT), which is basically an alternative Energy viewing gadget that enables automated alternative energy consumption from anywhere on the internet. We choose AT super controllers focusing mainly on mega controllers to comply with solar battery requirements. Our technology monitors the solar battery in real-time and transmits the capacity performance of the IoT system via the internet. We have a habit of using IOT Thing Speak to send alternate energy parameters to the IoT Thing Speak server's net. It now displays these metrics in a user-friendly graphical user interface and alerts the user when the performance limit is reached. This allows for easy remote observation of star plants while also ensuring optimal power production.

2. Literature survey

Aayushi "Arduino dependent solar monitoring gadget," given by Nitin Ingole at the 2014 International Conference on Science and Technology for Sustainable Development (ICSTSD). If the testing reveals a fault in the panels, the client will get the best performance from the panels. If your IoT sensors or other equipment develop problems that you don't have a way to verify, you'll have to depend on them [1]. Kabalci This article offers a rapid control infrastructure for a wind turbine and solar panel array-based sustainable energy production system. An unregulated six-pulse rectifier converts the DC turbine's output voltage to the AC output

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voltage of the wind turbine [2]. The charge voltage to the battery bank was provided by a hybrid system of renewable energy sources (RES) on the DC-bus, which generated voltages from wind turbines and solar panels. The current and voltage measurements of each renewable source are the focus of the tracking platform. The microchip's 18F4450 microcontroller calculates and interprets the appropriate values of the constructed sensor circuit. The circumstances under which they are being utilised the processed data is subsequently sent to a PC via USB, where it is stored in a database, and the device is monitored automatically. Viewable apps created using Microsoft Visual Studio. The web platform allows the system administrator to view the quantity of power generated and the current condition of each renewable energy source at a glance. The recorded data will be handled by the tracking software's programmed visual interface, which will assess the constant, weekly, and monthly values of each measurement individually.[3] Jiju K suggested the creation of Android-based online renewable energy monitoring and control systems. In this procedure, the Bluetooth interface of an Android tablet or phone is utilised as a data communication connection with the power conditioning device's digital hardware. Bluetooth's disadvantages include slower communication speeds, inadequate data security, and shorter battery life. The low-power design of Bluetooth limits the data transmission speed. H OhnishiAsH [4] demonstratedthe importance of the telecom sector in society is growing, as demonstrated by the exponential development of e-business and "dot-com" companies. In order to meet economic needs, the power-supply infrastructure utilised to provide utilities throughout this time period must be highly reliable as well as flexible. In this regard, one of the defining characteristics of contemporary networks is that traditional telecommunications systems produce more heat than networking systems. As a result, even if the power supply to telecommunications equipment is relatively steady, it is possible for a device to fail to owe to heat-related circumstances. As a consequence, air conditioning systems' power supplies, like those used by telecommunications equipment, must have excellent quality features. The goal of this article is to provide a technique for building a high-generation density data centre for telecommunications networks to achieve a high-reliability power supply in air conditioning systems.[5] Suzdalenko, Alexander in his article uses a non-intrusive load monitoring method to identify the problem of load disagreement in different devices when specific local generators based on renewable energy sources are linked to the same grid since they may be mismatched with the load changeable in time. Three methods for using electrical energy from locally based renewable energy sources are presented [6–10]. Moreover researchers are proposing various protocols in the field of healthcare [11–16] and vehicle communication [17–23] to protect the information exchanged among various devices to devices. Some researchers are providing various techniques for image privacy [24–28] and IoT based application [29–33].

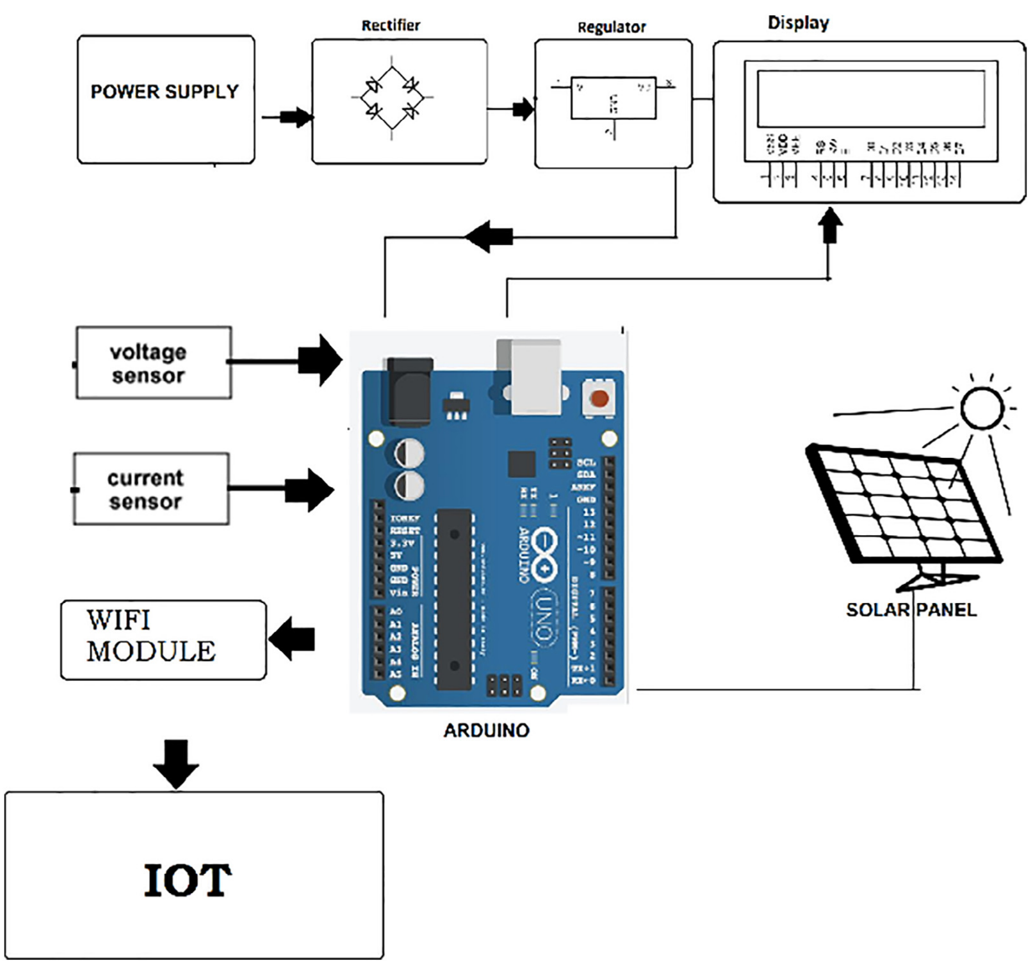

Fig. 1. Basic layout of the proposed system [2].

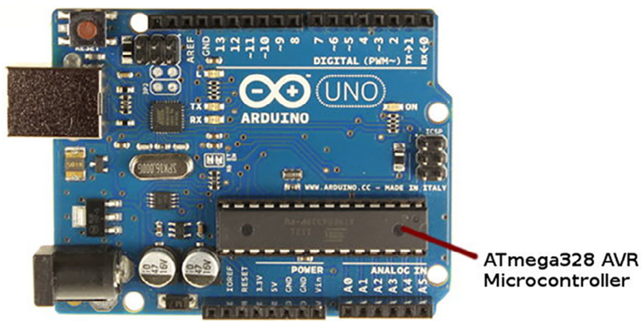

Fig. 2. ARDUINO Mega 328.

3. Proposed block diagram

See (Fig. 1.).

4. Important component description

4.1. Arduino

The ATmega328 is a microcontroller with Advanced Virtual RISC (AVR) capabilities. It can process data in 8 bits. The internal flash memory of the ATmega-328 is 32 KB. The ATmega328 features a 1 KB programmable read-only electrically erasable memory (EEPROM). This characteristic indicates that even if the microcontroller's electric source is disconnected, it can still store data and provide results after being reconnected to the power supply. Furthermore, the ATmega-328 features a 2 KB Static Random-Access Memory (SRAM) (SRAM). Other features will be discussed at a later time. The ATmega 328 offers a variety of characteristics that make it the most popular gadget on the market today. Advanced RISC architecture, excellent performance, low power consumption, real timer counter with an independent oscillator, 6 PWM pins, programmable Serial USART, programming lock for software security, throughput up to 20 MIPS, and so on are some of the characteristics.

AT Mega 328 is mostly used for its high versatility, along with its ease of use and familiarity. The AT Mega 328 connects solar panels and the Internet of Things (Internet of Things). For running, the AT Mega 328 needs a 5-volt DC supply. See (Fig. 2.).

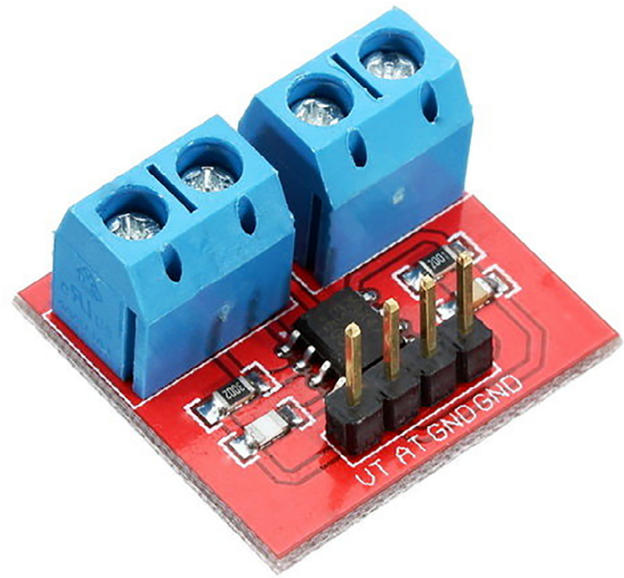

Fig. 4. Voltage and current sensor.

4.2. Solar PV panel

A solar panel, also known as a photovoltaic (PV) module, is an installation of photovoltaic cells placed in a framework. To produce electricity, solar cells absorb sunlight as a source of energy. A PV module transforms sunlight directly into direct current (DC) energy and is an important component of any PV system. PV mod-

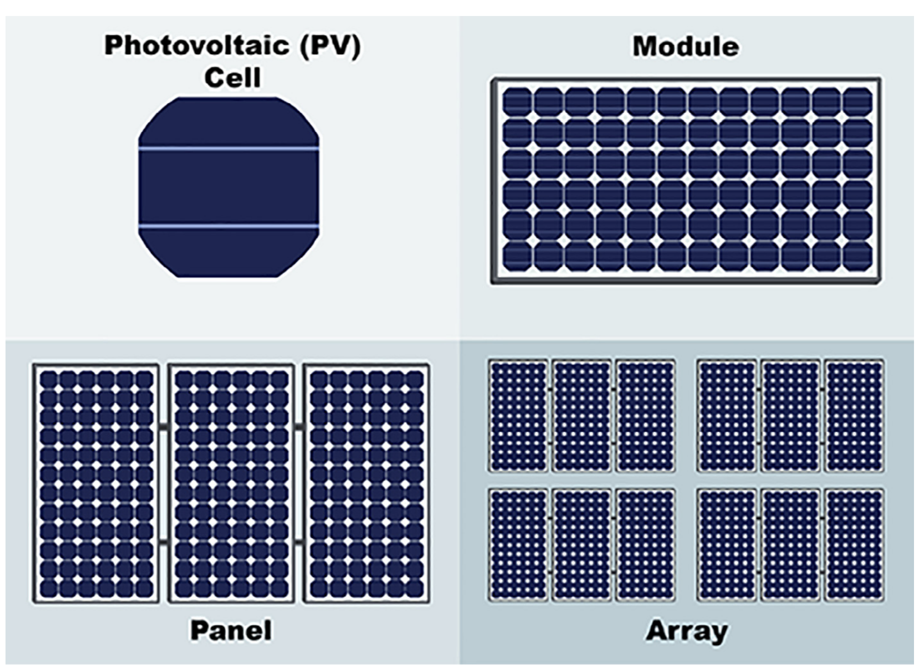

Fig. 3. Solar PV architecture hierarchy.

Fig. 5. Sample LCD display unit.

ules may be connected in series or parallel to provide the voltage and current that a given system needs. A solar module is made up of 6x10 solar cells in most cases. Depending on the kind and grade of solar cells utilised, the efficiency and wattage production may vary. A solar module's energy output may vary from 100 to 365 Watts of DC power. The greater the wattage output, the more energy each solar module is produced. As a result, a solar array of modules made up of higher-energy-producing solar modules would generate more power in less area than a solar array made up of lower-energy-producing solar modules. However, the cost of manufacturing more modules is higher. Although the monocrystalline solar panel is the original solar PV technology, it is being challenged on price, efficiency, and flexibility by both existing and developing new technologies. Residential, commercial, industrial, and utility customers now have a range of choices for meeting their solar energy production needs, thanks to polycrystalline silicon solar PV modules and new generations of thin-film solar PV technology. Depending on the needs of your project, the many solar power systems available range in efficiency, affordability, durability, and adaptability. The photovoltaic effect causes materials like silicon to produce an electrical current when they absorb sunlight, which is how PV solar technology generates electricity. Solar PV technology, like semiconductors, requires pure silicon to achieve maximum efficiency, and the crystalline silicon purification process typically determines the cost of PV solar production. A fundamental of solar PV architecture hierarchy has been shown in Fig. 3.

4.3. Voltage and current sensor

A voltage sensor is a device that measures and calculates the sum of voltage in an entity. Voltage sensors can detect AC or DC voltage levels, with voltage as the input and switches, analogue voltage signal, current signal, or audio signal as the output. As a current and power monitor, it measures the overall power used by the shunt load and sends the data to the AT super 328 in digital form. The latest reading of the shunt load is computed by the AT super 328 with the programmer loaded in it. Typical V&C sensor has been shown in Fig. 4.

4.4. LCD display

Liquid cell displays (LCDs) are used to depict numeric and alphanumeric characters in the matrix and segmental displays. They're found in notebook computers, digital clocks and watches, microwaves, CD players, and a range of other mobile devices throughout the US. LCDs are widely used because they have many benefits over other display technologies. Because they utilise interference light instead of transmitting it, LCDs consume a lot less electricity than junction rectifiers and gas-fired displays. An LCD | Liquid crystal display | Lcd | Digital display | Lcd | Liquid crystal display A passive or energetic matrix display grid may be used to create an alphanumeric display. Because each image element intersection in an energetic matrix includes a semiconductor unit,

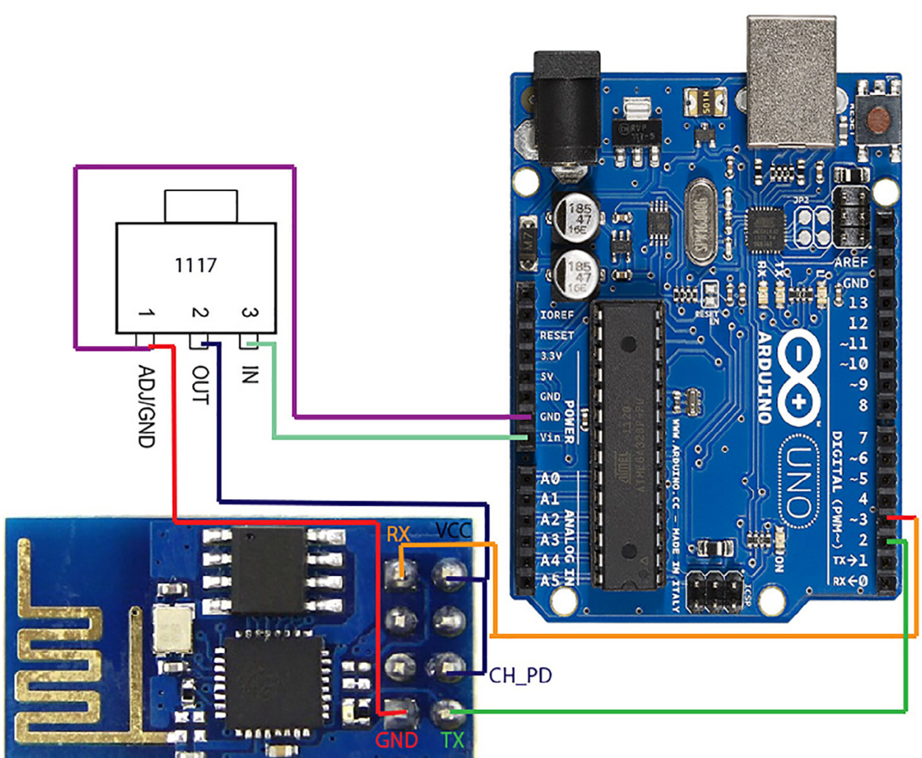

Fig. 6. WiFi Module.

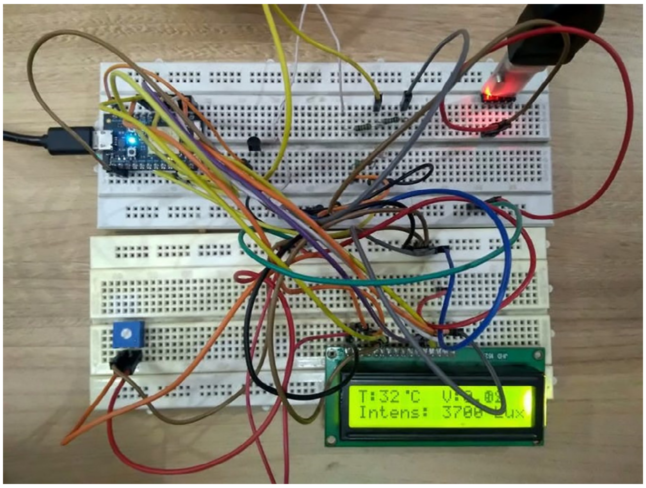

Fig. 7. Hardware assembly of IoT powered solar monitoring.

controlling the brightness of a picture element needs less current. As a consequence, the current may be switched on in an exciting matrix display while the refresh time on the screen is turned off. The passive matrix LCD has dual scanning, which means it scans the grid twice as quickly as the present model. See (Fig. 5. and Fig. 6.).

4.5. Wifi module

(ESP8266) The AT Mega 328 uses a Wi-Fi module to process all computed data before storing it on an IoT (Internet of Things) server or cloud. On a regular, weekly, and monthly basis, we analyse this data using the famous IoT platform Thing Speak.

5. Result & observation

The proposed system proves beneficial in multiple ways; Daily, weekly, and monthly analysis become easy and cost-effective since this technique maintains track of solar energy plants. Furthermore, any problem that happens within the powerhouse may be detected with the assistance of this research since the produced power may indicate some discrepancy in solar energy plant information. The observation clearly shows the improved performance of solar PV output. See (Fig. 7. and Fig. 8.).

6. Conclusion and future implications

Since the gadget needs a supply of 5 V and 3,3 V for service, this can be prevented only by using the solar array's energy. It is also feasible to follow the sun with the use of an engine and a primary for more excellent power production. Aside from that, it is possible to build a system clever enough to make choices regarding information and outcomes by integrating a range of Machine Learning algorithms and models. As the machine continues to monitor solar power plants, frequent, weekly, and monthly analysis becomes easy and trustworthy with the help of this study. Any fault in the power plant may be identified, and the generated power can reveal any discrepancies in data from solar power plants.

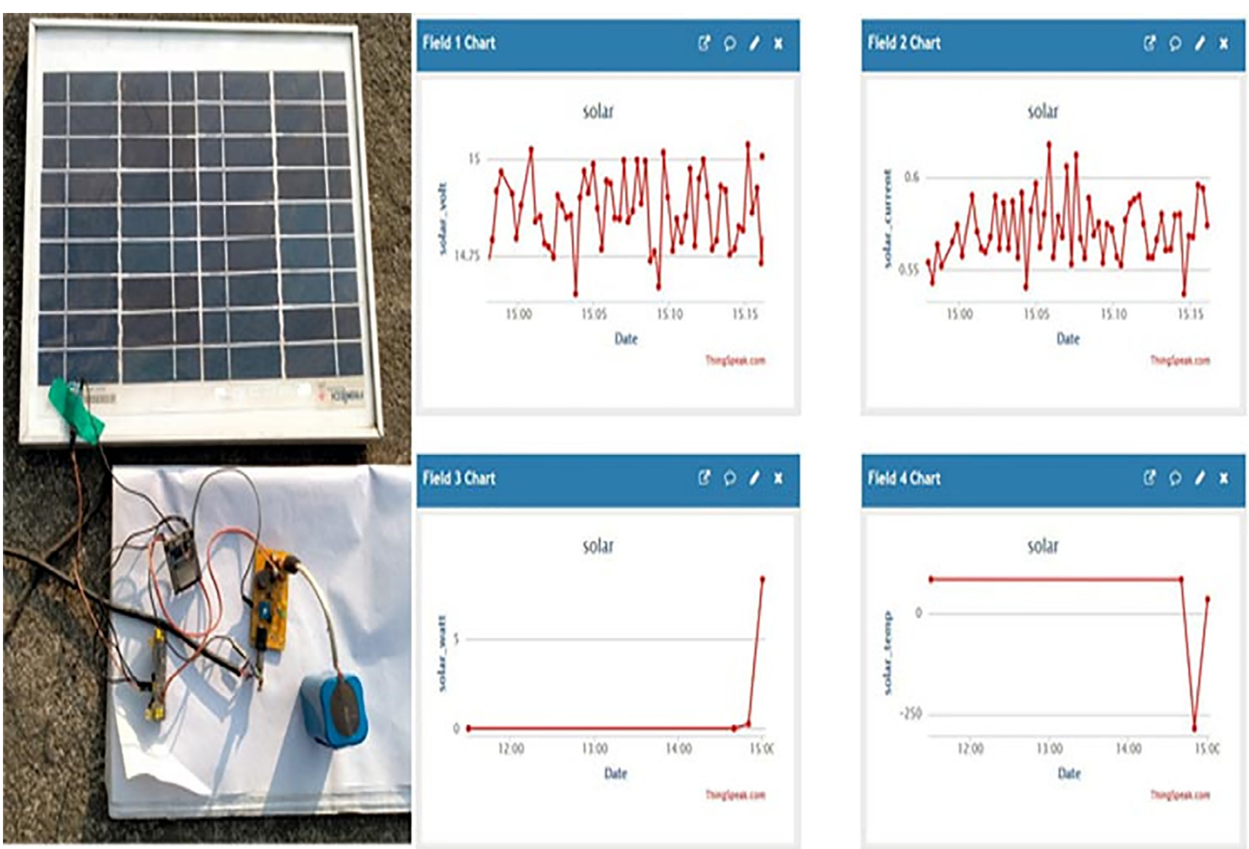

Fig. 8. Output Results.

CRediT authorship contribution statement

D. D. Prasanna Ran: Conceptualization, Data curation. **D. Suresh:** Formal analysis, Funding acquisition. **Prabhakara Rao Kapula:** Investigation, Methodology, Project administration. **C.H. Mohammad Akram:** Resources, Software, Supervisior. **N.Hemalatha:** Validation, Visualization, Writing. **Prem Kumar Soni:** Writing - original draft, Writing - review & editing.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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